

Chemistry, 2015 EC, entrance exam

Grade 12 · Chemistry · 2015 E.C.

Time allowed: 120 minutes

Questions: 76

Total marks: 76

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following is a colorless, flammable volatile liquid and miscible with non-polar solvents? (1)

A. Methane

B. Methanol

C. Ethanol

D. Benzene

2. Which one of the following petroleum products is used as a solvent and in dry cleaning? (1)

A. Diesel oil

B. Paraffin wax

C. Petroleum ether

D. kerosene

3. Which one of the following is NOT TRUE about laboratory preparation of ethylene by dehydration of ethanol? (1)

A. The collected ethylene gas in the experiment decolorizes alkaline KMnO_4 .

B. Combustion of ethylene produces a non-luminous flame.

C. Concentrated sulphuric acid is used as a dehydrating agent.

D. The collected ethylene gas in the experiment decolorizes Br_2 in CCl_4 .

4. Which one of the following is an essential nutrient of plants? (1)

A. Potassium

B. Sodium

C. Lead

D. Aluminum

5. In a kinetic experiment, a student placed crystals of iodine in a closed reaction vessel, introduced a given quantity of hydrogen gas, and collected data to calculate the rate of formation of hydrogen iodide. What would be the effect obtained on the rate of the reaction, if the student takes more amount of iodine from that used in the first experiment? (1)

A. The rate of the reaction will be constant.

B. The formation of hydrogen iodide will be decreased.

C. It lowers the rate of the reaction.

D. It speeds up the rate of the reaction.

6. Which of the following CORRECTLY relates the vapor pressure of a liquid with its boiling point? Boiling point is (1)

A. the temperature at which the kinetic energy of molecules is much less than the intermolecular force.

B. the temperature at which molecules present in the vapor phase go back to the liquid state.

C. the temperature at which the vapor pressure becomes equal to the external pressure.

D. the temperature at which all the three physical state co-exist.

7. Consider this equilibrium system: $\text{CO (g)} + \text{Fe}_3\text{O}_4 \text{ (s)} \rightleftharpoons \text{CO}_2 \text{ (g)} + 3 \text{FeO (s)}$ Which of the following disturbances shift the equilibrium position to the right? (1)

A. Adding $\text{Fe}_3\text{O}_4 \text{ (s)}$

B. Removing FeO (s)

C. Removing CO (g)

D. Removing $\text{CO}_2 \text{ (g)}$

8. Which one of the following resulted from interaction of the atomic orbitals of bonding atoms to produce new electron distribution associated with the entire molecule? (1)

- A. Bohr orbit** **B. Electron orbit**
C. Molecular orbital **D. Hybrid orbital**

9. Which type of organic substances would be produced by heating a mixture of a carboxylic acid and an alcohol in the presence of H_2SO_4 ?

- A. Ethers
B. Esters
C. Aldehydes
D. Ketones

10. The heat change at constant pressure that results when 1 mole of a compound is formed from its elements in their standard states is

- A. standard enthalpy of formation
B. standard reduction potential
C. standard enthalpy of reaction
D. standard entropy of reaction

11. What is a redox reaction? It is a reaction in which (1)

- A. an acid and a base neutralize each other.
- B. atoms or ions of two compound exchange bonding partners.
- C. electrons are transferred between species.
- D. two insoluble ions in compounds form an insoluble products.

12. Which one of the following explains how electroplating takes place? (1)

- A. Metal is deposited on the anode by reduction of ions in the electrolyte solution.
- B. The cations in the electrolyte are supplied either by the added salt or from oxidation of the anode.
- C. The object to be electroplated either metal or graphite-coated plastic constitutes the anode.
- D. The electrolyte is an aqueous solution of a salt of the plated metal.

13. How does the equivalent mass of a base is obtained? (1)

- A. By dividing its number of equivalent by the number of OH⁻ ions furnished by one formula unit of a base.
- B. By multiplying its number of equivalent by the number of OH⁻ ions furnished by one formula unit of a base.
- C. By dividing its molar mass by the number of OH⁻ ions furnished by one formula unit of a base.
- D. By multiplying its molar mass by the number of OH⁻ ions furnished by one formula unit of a base.

14. Which one of the following reactions would occur spontaneously in an aqueous solution at 25°C? Assume that the initial concentrations of the dissolved species are all 1.0 M. Half reaction $E^\circ(V)$ $\text{Sn}^{2+}(aq) + 2e^- \rightarrow \text{Sn}(s) -0.14$ $\text{Br}_2(l) + 2e^- \rightarrow 2\text{Br}^-(aq) +1.07$ $\text{Ag}^+(aq) + e^- \rightarrow \text{Ag}(s) +0.80$ $\text{Ni}^{2+}(aq) + 2e^- \rightarrow \text{Ni}(s) -0.25$ $\text{Ca}^{2+}(aq) + 2e^- \rightarrow \text{Ca}(s) -2.87$ $\text{Cd}^{2+}(aq) + 2e^- \rightarrow \text{Cd}(s) -0.40$ $\text{Cu}^{2+}(aq) + e^- \rightarrow \text{Cu}(s) +0.15$ $\text{Fe}^{3+}(aq) + e^- \rightarrow \text{Fe}^{2+}(aq) +0.77$

- A. $\text{Ca (s)} + \text{Cd}^{2+} \text{ (aq)} \rightarrow \text{Ca}^{2+} \text{ (aq)} + \text{Cd (s)}$
 B. $\text{Cu}^{2+} \text{ (aq)} + \text{Fe}^{2+} \text{ (aq)} \rightarrow \text{Cu (s)} + \text{Fe}^{3+} \text{ (aq)}$
 C. $2 \text{Br}^{-} \text{ (aq)} + \text{Sn}^{2+} \text{ (aq)} \rightarrow \text{Br}_2 \text{ (l)} + \text{Sn (s)}$
 D. $2 \text{Ag (s)} + \text{Ni}^{2+} \text{ (aq)} \rightarrow 2 \text{Ag}^{+} \text{ (aq)} + \text{Ni (s)}$

15. Which one of the following is the main reason for the large difference in dipole-moment between the triatomic molecules water (H_2O) and carbon dioxide (CO_2)? (1)
- A. The bonds in carbon dioxide (CO_2) are directed 180° and bond polarity reinforces each other.
B. The presence of multiple bonds in carbon dioxide (CO_2) while single bonds in water.
C. The higher electronegativity difference among atoms in water (H_2O) than in carbon dioxide (CO_2).
D. The presence of lone pairs of electrons in water (H_2O) partially reinforces bond polarities.
16. How many non-bonding pairs of electrons are present in the valence shell of the central atom in ClF_3 ? (1)
- A. 2
B. 3
C. 0
D. 1
17. Which one would be observed in a laboratory activity that is applied on naphthalene? (1)
- A. It is soluble in water.
B. It shows relatively high melting point.
C. It conducts electricity in molten state.
D. It is soluble in benzene.
18. Which one of the following statements is CORRECT? (1)
- A. Valence bond model explain molecular shape from the interaction of atomic orbitals.
B. In Lewis model, electrons repel one another and a pair of bonding electrons lies far apart as possible.
C. Lewis model predicts the shapes, of molecules based on electron-pair repulsions around a central atom.
D. Valence bond model simply shows the linkage between atoms and the presence of lone pairs.
19. The energy required to completely separate one mole of a solid ionic compound into gaseous ions is a/an (1)
- A. Bond energy
B. Lattice energy
C. Chemical potential
D. Boiling Point
20. Which of the properties of solvents WOULD NOT be affected by the formation of a solution with a non-volatile solute? (1)
- A. Freezing Point
B. Chemical identity
C. Chemical potential
D. Boiling Point
21. The partial pressure of acetylene, C_2H_2 , in a container is 10 atm at 25°C . What is the solubility in molarity of C_2H_2 , (1) assume that C_2H_2 obeys Henry's law? ($K_H = 1.04 \text{ mol/L atm}$)
- A. 10.4
B. 104
C. 0.104
D. 9.62
22. What is the normality of a solution that contains 19.6 g of H_2SO_4 in 20 mL of the solution? (1)
- A. 10
B. 20
C. 0.01
D. 0.02
23. Which one of the following illustrates an extensive property? (1)
- A. If water in two beakers at the same temperature are mixed to make a single quantity of water and the temperature of the combined quantity of water is the same as it was in the two separate beakers.
B. Doubling the volume of a sample also doubles the mass; so the ratio of mass to volume remains the same.
C. If a system is divided into parts and it is found that the property of the complete system has a value that is the sum of the values of the property of all the parts.
D. The electrode potential at standard condition for the half-reaction $2 \text{Cu}^{2+} + 4 \text{e}^- \rightarrow 2 \text{Cu}$ is the same as for $\text{Cu}^{2+} + 2 \text{e}^- \rightarrow \text{Cu}$.

-
- 24.** The point in a titration at which the number of moles of OH^- ions added to a solution is equal to the number of moles of H^+ ions originally present is (1)
- A. Equivalence point
B. Triple point
C. End point
D. Zero point
-
- 25.** Which of the following statements describes chemical industries? It is (1)
- A. a firm that involves usage of raw material to produce desired products by using chemical reactions and refining methods.
B. a process in which dissolved solute comes out of solution and forms crystals.
C. an expression for the chemical composition of a compound in terms of the symbols of the atoms of the elements involved.
D. an area of chemistry concerned with the speeds, or rates, at which chemical reactions occur.
-
- 26.** What will be the product if phosphorus reacts with excess amount of oxygen? (1)
- A. P_4O_6
B. P_4O_{10}
C. H_3PO_4
D. H_3PO_3
-
- 27.** One of the basic differences between metallic conduction and electrolytic conduction is: (1)
- A. The conduction of electricity is due to the movement of ions in metallic conduction while the flow of mobile electrons in electrolytic conduction.
B. The charge carriers in metals are localized electrons while strongly held ions are in electrolytic conduction.
C. The charge carriers in metals are cations and anions while mobile electrons are in electrolytic conduction.
D. The conduction of electricity is due to the movement of ions in an electrolytic conduction while the flow of mobile electrons in metallic conduction.
-
- 28.** Which one of the following gases is released from combustion of a fuel in an industrial machinery causing global warming? (1)
- A. SO_2
B. NO
C. CO_2
D. CO
-
- 29.** The number of digits to the right of the decimal point denotes: (1)
- A. Scientific notation
B. Conversion factor
C. Decimal places
D. Significant figures
-
- 30.** Which one of the following IS TRUE about isotopes? (1)
- A. Isotopes of an element have the same chemical properties.
B. Isotopes of the same elements contain the same number of neutrons.
C. Isotopes of an element are identical in mass number.
D. Isotopes of an element are atoms that have different number of electrons and protons.
-
- 31.** Which one of the following describes the quantum mechanical model? (1)
- A. Electrons moved around the nucleus of an atom in circular orbits.
B. An atom is a sphere with a positively charged matter, in which electrons are embedded.
C. An atom is represented by a solid indestructible sphere.
D. The locations of electrons in an atom are described in terms of probability.

32. Which one of the following is an ionic compound? (1)

- A. H₂O₂
B. PCl₃
C. CO₂
D. CaCl₂

33. An element is located in Group VIIA and 3rd period of the modern periodic table. On the basis of this the element;

- A. has extremely low electron affinity.
- B. forms an acidic oxide.
- C. has very low ionization energy.
- D. generally forms an ion with -2 charges.

34. Which one of the following is the reducing agent in the following reaction? $8\text{NH}_3(\text{g}) + 6\text{NO}_2(\text{g}) \rightarrow 7\text{N}_2(\text{g}) + 12\text{H}_2\text{O}(\text{l})$ (1)

- A. N₂
B. H₂O
C. NH₃
D. NO₂

35. Carbon forms two stable compounds with oxygen, namely, carbon monoxide and carbon dioxide. For a fixed mass of carbon, the ratio of oxygen in carbon monoxide to oxygen in carbon dioxide is 1:2. This result is consistent with the law of

- A.** multiple proportion.
B. constant composition.
C. conservation of mass.
D. definite proportion.

36. When a system at equilibrium is disturbed by a change of variable, the system shifts the equilibrium position in a way that tends to counteract this change of variable. This is a statement for

- A. Le Chatelier Principle
B. Heisenberg uncertainty principle
C. Aufbau principle
D. Pauli's exclusion principle

37. Which one of the following gas laws states the volume of a fixed amount of gas maintained at constant pressure is directly proportional to the absolute temperature of the gas? (1)

- A. Combined gas law
B. Avogadro's law
C. Boyle's law
D. Charles' law

38. Which one of the following is an example of molecular crystals? (1)

- A.** Magnesium **B.** Solid carbon dioxide
C. Sodium Chloride **D.** Diamond

39. Which one of the following COULD NOT be considered as factor influencing the rate of chemical reaction? (1)

- A. Temperature of reactants
B. The rate constant of the reaction
C. Surface area of reactants
D. Presence of catalyst

40. In which of the following order of a reaction, a plot of $\ln [A]_t$ versus time is a straight line whose slope is $-k$? (1)

- A. Second
B. Third
C. Zero
D. First

41. A forward and reverse reaction continues to take place at equal rate in microscopic level without change in composition at macroscopic level. This statement is the definition of (1)

- A. The law of mass action
B. Degree of freedom
C. Dynamic equilibrium
D. Chemical kinetics

42. What are the four quantum numbers for an electron in a 2s orbital? (1)

- A. $2,0,1,+1/2$ B. $2,0,0,+1/2$
C. $2,1,1,+1/2$ D. $2,1,0,+1/2$

43. A student uses a digital balance (± 0.02), a thermometer (± 0.1), a measuring cylinder (± 0.05), and a beaker (± 0.5) during an experiment. Which of the above instrument is more precise?

- A.** The measuring cylinder **B.** The Beaker
C. The digital balance **D.** The thermometer

44. Which one of the following explains that light has BOTH wave and particle nature? (1)

- A.** Light consists of electromagnetic waves which have an oscillating electric and magnetic disturbance that spreads electromagnetic radiation.
- B.** Microscopic systems, such as atoms and molecules can possess wave properties.
- C.** An electron that is bound to the nucleus behaves like a standing wave and its position is given in terms of probability.
- D.** A beam of light is a stream of particles of electromagnetic energy, with energy proportional to the observed frequency of the light.

45. A nuclear breakdown in which particles or electromagnetic radiation is emitted is _____.

- A.** Radioactive decay **B.** Radio wave
- C.** Radioactivity **D.** Radioactive isotopes

46. Which statement describes valence shell electron pair repulsion (VSEPR) theory? (1)

- A. A covalent bond is formed when orbitals of two atoms overlap and the overlap region is occupied by a pair of electrons.
- B. Atomic orbitals are capable of combining to produce new electron distribution.
- C. Electron pairs will be as far apart from each other in three dimensional spaces as possible.
- D. Each degenerate orbitals are occupied by a single electron before the second electrons of opposite spin enters the orbitals.

47. Which of the following compounds reacts with basic oxide to form salt and water? (1)

- A.** Neutral oxide **B.** Base
C. Acidic oxide **D.** Acid

48. Which one of the following reaction represents formation of an acid from its salt? (1)

- A. $\text{AgNO}_3 (\text{aq}) + \text{NaCl} (\text{aq}) \rightarrow \text{NaNO}_3 (\text{aq}) + \text{AgCl} (\text{s})$
 B. $\text{NaNO}_3 (\text{s}) + \text{H}_2\text{SO}_4 (\text{l}) \rightarrow \text{NaHSO}_4 (\text{s}) + \text{HNO}_3 (\text{l})$
 C. $\text{P}_2\text{O}_5 (\text{s}) + 6 \text{NaOH} (\text{aq}) \rightarrow 2 \text{Na}_3\text{PO}_4 (\text{s}) + 3 \text{H}_2\text{O} (\text{l})$
 D. $\text{CH}_3\text{COOH} (\text{aq}) + \text{CH}_3\text{OH} (\text{aq}) \rightarrow \text{CH}_3\text{CO}_2\text{CH}_3 (\text{aq}) + \text{H}_2\text{O} (\text{l})$

49. The negative logarithm of the molar concentration of hydrogen ion to the base ten is known as _____.

- A. pOH
B. K_a
C. K_m
D. pH

50. Which one of the following galvanic cell is an example of secondary cell? (1)

- A. Zinc-carbon dry cell
B. Leclanche dry cell
C. Lead storage cell
D. Daniel's cell

51. Which one of the following is responsible for electrolytic conductivity? (1)

- A.** Movement of ions.
B. Electrons held in a fixed position.
C. Free electrons.
D. Ions strongly held in a fixed position.

52. The pH of a 0.1 M solution of formic acid (HCOOH) is 3. What is the K_a of the acid? (1)

- A. 9.99×10^{-5} B. 9.99×10^{-4}
C. 1.01×10^{-7} D. 1.01×10^{-5}

53. A system does 15 J of work and no other changes take place. Which of the following function has fallen by 15 J? (1)

- A. Entropy
B. Enthalpy
C. Specific heat
D. Internal energy

54. Which of the following is CORRECTLY stated about the spontaneity of a reaction? (1)

- A. For an exothermic reaction with negative reaction entropy, the reaction is spontaneous at high temperature.
- B. For an endothermic reaction with positive reaction entropy, the reaction is non-spontaneous at low temperature.
- C. For an exothermic reaction with positive reaction entropy, the reaction is non-spontaneous at all temperature.
- D. For an endothermic reaction with negative reaction entropy, the reaction is spontaneous at all temperature.

55. In thermodynamics: solutions in a beaker, a gas in a cylinder, a reaction vessel, a biological cell are examples of (1)

- A. thermodynamic system
B. path function
C. state function
D. thermodynamic variables

56. Given the following: bond energies (kJ/mol) H-H 936 N-N 193 N=N 409 N≡N 942 and the enthalpy for the reaction $1/2 \text{N}_2(\text{g}) + 3/2 \text{H}_2(\text{g}) \rightarrow \text{NH}_3(\text{g})$; $\Delta H = -45 \text{ kJ/mol}$ What is the bond energy for NH_3 ? (1)

- A. 1080
B. 960
C. 1170
D. 1125

57. Which of the following would occur during the formation of ions? (1)

- A. Valence electrons are transferred from one atom to another.
- B. The electronic configuration of an atom forming the ion remains unaffected.
- C. Valence electrons are shared between atoms.
- D. Inner most shell electrons are involved.

58. In order to illustrate the structure and properties of a solid, a student put hundred marble balls in to a box that representing metal cations and the empty space between and around the marble representing the electron sea. Which one of the following bonding is represented by this model? (1)

- A.** Dative bonding **B.** Metallic bonding
C. Ionic bonding **D.** Covalent bonding

59. Which one of the following explains dipole-dipole force? (1)

- A. An electrostatic attraction between metal ions and valence electrons.
B. An attractive force resulting from temporary dipoles induced in molecules.
C. The intermolecular attraction between oppositely charged poles of nearby polar molecules.
D. A type of intermolecular force that exists between non- polar molecules.

60. Which one of the following reactions creates energy in our bodies during respiration? (1)
- A. $\text{N}_2\text{H}_4\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightarrow 2\text{NH}_3(\text{l}) + \text{CO}_2(\text{g})$
B. $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) \rightarrow 2\text{C}_2\text{H}_5\text{OH}(\text{l}) + 2\text{CO}_2(\text{g})$
C. $6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6\text{O}_2(\text{g})$
D. $\text{C}_6\text{H}_{12}\text{O}_6(\text{s}) + 6\text{O}_2(\text{g}) \rightarrow 6\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$
61. "A reaction that involves the exchange of positive and negative ions of each reactant". This is a definition of a (1)
- A. Single displacement reaction
B. Double displacement reaction
C. Combination reaction
D. Decomposition reaction
62. Which one of the following statement is TRUE about gases? (1)
- A. The volume of a gas is equal to the volume of its container.
B. When a gas sample is heated at constant pressure, its volume decreases.
C. The volume of a gas is kept constant as pressure increases.
D. The particles of gases are closely packed together.
63. Which one of the following mathematical expressions is derived from the combination of Boyle's law, Charles' law and Avogadro's law? (1)
- A. $V \propto (PnT)$
B. $V \propto (nP T)$
C. $V \propto (Tn P)$
D. $V \propto (nT P)$
64. 180 g of calcium carbonate (CaCO_3) is allowed to react with 156 gram of hydrochloric acid (HCl), according to the following reaction. Which of the following substances is the limiting reagent? (Atomic masses; Ca=40, Cl 35.5, O=16, C=12, H=1) $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$ (1)
- A. CO_2
B. H_2O
C. CaCO_3
D. HCl
65. Which one of the following isomers differs in the arrangement of carbon atoms in the longest continuous carbon chain? (1)
- A. Structural
B. Cis-trans
C. Position
D. Geometric
66. Which one of the following methods can be used in the laboratory for the preparation of alkynes? (1)
- A. Hydrogenation of alkenes with a metal catalyst.
B. Dehydrohalogenation of adjacent dihalides with sodium hydroxide.
C. Heating of sodium salt of an organic acid with soda lime.
D. Dehydration of alcohols with concentrated sulphuric acid.
67. Which one of the following is TRUE about the electrolysis of a molten hypothetical electrolyte, MX? (1)
- A. The overall reaction is redox reaction.
B. Electrons are not transferred from one species to the other.
C. M is produced at the anode.
D. X is produced at the negative electrode.

68. During the extraction of an unknown metal, the metallic ore was fused with NaOH in air and then dissolved in water to obtain the soluble unknown metal containing species. Reduction with carbon is followed after the solution is treated with HCl. Finally the obtained oxide is reduced to its metallic form by aluminum. The most probable unknown metal would be (1)
- A. Pb
B. Cr
C. Zn
D. Ca

69. Why tin does NOT react with cold water but calcium reacts with cold water quite readily? (1)
- A. Due to the pH of water.
B. Due to its physical state.
C. Due to the formation of a thin oxide layer.
D. Due to its less reactive nature.

70. Why is vulcanization performed in natural rubber? (1)
- A. To increase the number of monomer unit in the natural rubber.
B. To decrease the number of monomer unit in the natural rubber.
C. To establish cross-links between polymer chain in the natural rubber.
D. To convert the monomer unit in natural rubber in to polymer.

71. Which one of the following explains the determination of the cell potential of Zn-Cu cell using voltmeter? (1)
- A. After the experiment the mass of zinc is increased.
B. Each U-tube served as a salt bridge that can be filled with saturated KCl.
C. In the Zn half- cell, Zn metal will be dipped in CuSO_4 solution.
D. In the Cu half-cell, Cu metal will be dipped in ZnSO_4 solution.

72. The rate law for: is rate= . The accepted mechanism for the reaction is Why DOES NOT H_2O appear in the rate law? Beacuse (1)
- A. water is missing in the rate determining step.
B. water exists in the liquid state in the reaction.
C. water is an intermediate species.
D. water is a catalyst in the reaction.

73. Which type of substances has the following general formula? (R_1 , R_2 and R_3 hydrocarbon groups.) (1)
- A. Fatty acids
B. Alcohols
C. Fats
D. Carboxylic acids

74. The rate law for this reaction $\text{CO}(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{NO}(\text{g})$ is rate = $k[\text{NO}_2]^2$. Which one of the following could be a plausible mechanism for the reaction? (1)
- A. $\text{NO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{NO}_3(\text{g}) + \text{NO}_2(\text{g})$ [fast] $\text{NO}_3(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{CO}_2(\text{g})$ [slow]
B. $\text{NO}_3(\text{g}) + \text{NO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}_2(\text{g})$ [slow] $\text{NO}_2(\text{g}) + \text{CO}_2(\text{g}) \rightarrow \text{NO}_3(\text{g}) + \text{CO}_2(\text{g})$ [fast]
C. $\text{NO}_3(\text{g}) + \text{NO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{NO}_2(\text{g})$ [slow] $\text{NO}_3(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{CO}_2(\text{g})$ [fast]
D. $\text{NO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{NO}_3(\text{g}) + \text{NO}(\text{g})$ [slow] $\text{NO}_3(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}_2(\text{g}) + \text{CO}_2(\text{g})$ [fast]

75. The speed with which a solute goes into solution is (1)
- A. extent of dissolution
B. rate of dissolution
C. heat of solution
D. degree of solvation

76. Which one of the following explains how 2 L of 1 M NaOH aqueous stock solution is converted into a 0.25 M NaOH solution?

(1)

- A. Evaporating 1 L of water from the stock solution
- B. Evaporating 0.25 L of water from the stock solution
- C. Adding 8 L water to the stock solution
- D. Adding 6 L water to the stock solution

— END OF EXAMINATION —

Generated on Meskerem 21, 2019 E.C. (October 1, 2026 G.C.) · Temari.et